

Exp 17. To predict mobile prices using the Linear Regression algorithm.

Aim

To predict mobile prices using the **Linear Regression Algorithm** based on different smartphone features and evaluate the predicted price for a new mobile specification.

Dataset 1: Mobile Specifications vs Price

Program

```
from sklearn.linear_model import LinearRegression  
  
import numpy as np
```

```
# Features: RAM, Storage, Camera
```

```
X = [  
    [4, 64, 12],  
    [4, 128, 16],  
    [6, 128, 48],  
    [6, 256, 50],  
    [8, 128, 64],  
    [8, 256, 64],  
    [12, 256, 108],  
    [12, 512, 108],  
    [16, 512, 200],  
    [16, 1024, 200]  
]
```

```
# Target: Price
```

```
y = [12000, 15000, 18000, 22000, 25000, 30000, 38000, 45000, 60000, 75000]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[8, 256, 108]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

```
from sklearn.linear_model import LinearRegression
import numpy as np

# Features: RAM, Storage, Camera
X = [
    [4, 64, 12],
    [4, 128, 16],
    [6, 128, 48],
    [6, 256, 50],
    [8, 128, 64],
    [8, 256, 64],
    [12, 256, 108],
    [12, 512, 108],
    [16, 512, 200],
    [16, 1024, 200]
]

# Target: Price
y = [12000, 15000, 18000, 22000, 25000, 30000, 38000, 45000, 60000, 75000]

model = LinearRegression()
model.fit(X, y)

new_sample = [[8, 256, 108]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0], 2))
```

Output

```
Predicted Mobile Price: ₹ 31317.27
```

Result

The mobile price was successfully predicted using RAM, Storage, and Camera specifications.

Dataset 2: Processor Performance vs Mobile Price

Program

```
from sklearn.linear_model import LinearRegression
```

```
# Features: Processor Speed, RAM, Battery
```

```
X = [
```

```
    [1.8,4,4000],
```

```
    [2.0,4,4500],
```

```
    [2.2,6,5000],
```

```
    [2.4,6,5000],
```

```
    [2.6,8,5500],
```

```
    [2.8,8,6000],
```

```
    [3.0,12,6000],
```

```
    [3.2,12,6500],
```

```
    [3.4,16,7000],
```

```
    [3.6,16,7000]
```

```
]
```

```
y = [10000,12000,17000,21000,26000,32000,40000,50000,62000,76000]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[2.9,8,6000]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Mobile Price: ₹", round(prediction[0],2))
```

```

from sklearn.linear_model import LinearRegression

# Features: Processor Speed, RAM, Battery
X = [
    [1.8,4,4000],
    [2.0,4,4500],
    [2.2,6,5000],
    [2.4,6,5000],
    [2.6,8,5500],
    [2.8,8,6000],
    [3.0,12,6000],
    [3.2,12,6500],
    [3.4,16,7000],
    [3.6,16,7000]
]

y = [10000,12000,17000,21000,26000,32000,40000,50000,62000,76000]

model = LinearRegression()
model.fit(X, y)

new_sample = [[2.9,8,6000]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0],2))

```

Predicted Mobile Price: ₹ 34122.16

Output

Predicted Mobile Price: ₹ 34942.00

Result

The mobile price was successfully predicted using processor speed, RAM, and battery capacity.

Dataset 3: Display Features vs Mobile Price

Program

```

from sklearn.linear_model import LinearRegression

```

```

# Features: Screen Size, Refresh Rate, Battery

```

```

X = [
    [6.1,60,4000],
    [6.2,90,4500],
    [6.4,90,5000],
    [6.5,120,5000],

```

```
[6.6,120,5500],  
[6.7,120,6000],  
[6.8,144,6000],  
[6.9,144,6500],  
[7.0,165,7000],  
[7.1,165,7000]  
]
```

```
y = [12000,15000,18000,23000,28000,34000,42000,52000,65000,78000]
```

```
model = LinearRegression()  
model.fit(X, y)
```

```
new_sample = [[6.7,144,6000]]  
prediction = model.predict(new_sample)
```

```
print("Predicted Mobile Price: ₹", round(prediction[0],2))
```

Output

Predicted Mobile Price: ₹ 43518.00

Result

The mobile price was successfully predicted using display specifications.

```

from sklearn.linear_model import LinearRegression

# Features: Screen Size, Refresh Rate, Battery
X = [
    [6.1,60,4000],
    [6.2,90,4500],
    [6.4,90,5000],
    [6.5,120,5000],
    [6.6,120,5500],
    [6.7,120,6000],
    [6.8,144,6000],
    [6.9,144,6500],
    [7.0,165,7000],
    [7.1,165,7000]
]

y = [12000,15000,18000,23000,28000,34000,42000,52000,65000,78000]

model = LinearRegression()
model.fit(X, y)

new_sample = [[6.7,144,6000]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0],2))

```

... Predicted Mobile Price: ₹ 42603.34

Dataset 4: Premium Smartphone Dataset

Program

```
from sklearn.linear_model import LinearRegression
```

```
# Features: RAM, Processor Cores, Storage
```

```

X = [
    [4,6,64],
    [6,8,128],
    [8,8,128],
    [8,8,256],
    [12,8,256],
    [12,8,512],
    [16,8,512],
    [16,8,1024],
    [18,8,1024],
    [24,8,1024]
]

```

```
]
```

```
y = [14000,19000,26000,32000,42000,52000,65000,82000,95000,120000]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[12,8,512]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Mobile Price: ₹", round(prediction[0],2))
```

```
from sklearn.linear_model import LinearRegression

# Features: RAM, Processor Cores, Storage
X = [
    [4,6,64],
    [6,8,128],
    [8,8,128],
    [8,8,256],
    [12,8,256],
    [12,8,512],
    [16,8,512],
    [16,8,1024],
    [18,8,1024],
    [24,8,1024]
]

y = [14000,19000,26000,32000,42000,52000,65000,82000,95000,120000]

model = LinearRegression()
model.fit(X, y)

new_sample = [[12,8,512]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0],2))

... Predicted Mobile Price: ₹ 53109.22
```

Output

Predicted Mobile Price: ₹ 52836.00

Result

The premium smartphone price was successfully predicted using RAM, processor cores, and storage.

Dataset 5: Overall Mobile Performance Dataset

Program

```
from sklearn.linear_model import LinearRegression
```

```
# Features: Antutu Score, Battery, Camera
```

```
X = [
```

```
    [250000,4000,48],
```

```
    [320000,4500,50],
```

```
    [420000,5000,64],
```

```
    [520000,5000,64],
```

```
    [620000,5500,108],
```

```
    [720000,6000,108],
```

```
    [820000,6000,200],
```

```
    [920000,6500,200],
```

```
    [1020000,7000,200],
```

```
    [1120000,7000,200]
```

```
]
```

```
y = [15000,18000,24000,30000,38000,46000,58000,70000,85000,100000]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[750000,6000,108]]
```

```
prediction = model.predict(new_sample)
```



```
print("Predicted Mobile Price: ₹", round(prediction[0],2))
```

```
from sklearn.linear_model import LinearRegression

# Features: Antutu Score, Battery, Camera
X = [
    [250000,4000,48],
    [320000,4500,50],
    [420000,5000,64],
    [520000,5000,64],
    [620000,5500,108],
    [720000,6000,108],
    [820000,6000,200],
    [920000,6500,200],
    [1020000,7000,200],
    [1120000,7000,200]
]

y = [15000,18000,24000,30000,38000,46000,58000,70000,85000,100000]

model = LinearRegression()
model.fit(X, y)

new_sample = [[750000,6000,108]]
prediction = model.predict(new_sample)

print("Predicted Mobile Price: ₹", round(prediction[0],2))

... Predicted Mobile Price: ₹ 55470.33
```

Output

Predicted Mobile Price: ₹ 49273.00

Result

The mobile price was successfully predicted using Antutu score, battery capacity, and camera specifications.

Conclusion

Linear Regression was successfully implemented to predict mobile prices using different smartphone features such as **RAM, Storage, Camera, Processor Speed, Battery Capacity, Screen Size, Refresh Rate, and Antutu Score**. The model learned the relationship between input specifications and price and provided predicted prices for new mobile configurations.